

Rethinking Soundness: From Local Rules to Global Correctness

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In research...

... we rarely begin with a polished proof.

- ▶ Conjectures,
- ▶ sketches,
- ▶ examples,
- ▶ and intuitions – many of which turn out to be incomplete or even wrong.

Not every intermediate thought is correct, but the overall process eventually subjects those ideas to sufficient structure: formalization, proof, discussion with colleagues, peer review.

In research...

many breakthroughs started as not fully justified steps.

- ▶ heuristics in SAT/SMT solving: modern solvers guess which variable to assign next, restart the search, etc.
- ▶ abstraction refinement: methods that intentionally ignore information to make a problem tractable.
- ▶ in mathematics: exploratory reasoning precedes formalization.

Research often starts in a space where "this is not yet a proof" is nothing unusual.

Case study: proof theory

A proof is a sequence of formulas $A_1, \dots, A_n$, each of which is either an axiom or follows by direct rule application to formulas with lower index.

- ▶ Every inference rule should be **locally sound**: whenever the premises are true, so is the conclusion.

This locality is so natural that we often forget it is **an assumption** rather than a law of nature.

Case study: proof theory

There are interesting situations in which this assumption gets in the way.

- ▶ Logical systems, where insisting on locally sound rules prevents us from obtaining a satisfactory proof system.
- ▶ We can sometimes recover soundness globally, even though individual inference steps are not sound in isolation.

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Sequent calculus **LK**: $\Gamma \vdash \Delta \quad \bigwedge \Gamma \rightarrow \bigvee \Delta$

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$$\frac{\Gamma \vdash \Delta}{A, \Gamma \vdash \Delta} w_l$$

$$\frac{\Gamma \vdash \Delta}{\Gamma \vdash \Delta, A} w_r$$

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cut-elimination theorem

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$$\frac{A, B, \Gamma \vdash \Delta}{A \wedge B, \Gamma \vdash \Delta} \wedge_l$$

$$\frac{\Gamma \vdash \Delta, A \quad \Gamma \vdash \Delta, B}{\Gamma \vdash \Delta, A \wedge B} \wedge_r$$

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$$\frac{\Gamma \vdash \Delta, A, B}{\Gamma \vdash \Delta, A \vee B} \vee_r$$

$$\frac{A, \Gamma \vdash \Delta \quad B, \Gamma \vdash \Delta}{A \vee B, \Gamma \vdash \Delta} \vee_l$$

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$$\frac{A, \Gamma \vdash \Delta, B}{\Gamma \vdash \Delta, A \rightarrow B} \rightarrow_r$$

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$$\frac{\Gamma \vdash \Delta, A}{\neg A, \Gamma \vdash \Delta} \neg_l$$

$$\frac{A, \Gamma \vdash \Delta}{\Gamma \vdash \Delta, \neg A} \neg_r$$

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Sequent calculus **LK**: $\Gamma \vdash \Delta \quad \wedge \Gamma \rightarrow \vee \Delta$

$$\frac{A(t), \Gamma \vdash \Delta}{\forall x A(x), \Gamma \vdash \Delta} \forall_l$$

$$\frac{\Gamma \vdash \Delta, A(a)}{\Gamma \vdash \Delta, \forall x A(x)} \forall_r$$

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$$\frac{A(t), \Gamma \vdash \Delta}{\forall x A(x), \Gamma \vdash \Delta} \forall_I$$

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a is an **eigenvariable** $\rightarrow$ **regularity**

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LK is sound and complete, and the cut-elimination theorem holds.

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How are inference rules formed?

Macros for connectives

A macro for connectives is a formula based on propositional variables which is considered as a connective in its own right.

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Example

The binary connective $\leftrightarrow$:

$$A \leftrightarrow B = (A \rightarrow B) \wedge (B \rightarrow A).$$

Macros for connectives

$$A \leftrightarrow B = (A \rightarrow B) \wedge (B \rightarrow A)$$

$\leftrightarrow_r$ is defined via

$$\frac{\frac{A \vdash B}{\vdash A \rightarrow B} \rightarrow_r \quad \frac{B \vdash A}{\vdash B \rightarrow A} \rightarrow_r}{\vdash (A \rightarrow B) \wedge (B \rightarrow A)} \wedge_r$$

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Consequently, the inference rule for $\leftrightarrow$ is

$$\frac{A, \Gamma \vdash \Delta, B \quad B, \Gamma \vdash \Delta, A}{\Gamma \vdash \Delta, A \leftrightarrow B} \leftrightarrow_r$$

Macros for connectives

$\leftrightarrow_I$ is defined via

$$\frac{\frac{\frac{\vdash A, B}{B \rightarrow A \vdash A} \rightarrow_I \quad \frac{A \vdash A}{A, B \vdash} \rightarrow_I}{B, B \rightarrow A \vdash} \rightarrow_I \quad \frac{B \vdash B \quad A, B \vdash}{(A \rightarrow B), (B \rightarrow A) \vdash} \rightarrow_I}{(A \rightarrow B) \wedge (B \rightarrow A) \vdash} \wedge_I$$

Macros for connectives

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$$\frac{\frac{\frac{\vdash A, B}{B \rightarrow A \vdash A} \rightarrow_I \quad \frac{A \vdash A}{A, B \vdash} \rightarrow_I}{(A \rightarrow B), (B \rightarrow A) \vdash} \rightarrow_I \quad \frac{(A \rightarrow B), (B \rightarrow A) \vdash}{(A \rightarrow B) \wedge (B \rightarrow A) \vdash} \wedge_I$$

Consequently, the inference rule for $\leftrightarrow$ is

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Macros for connectives

LK is extended with

$$\frac{A, \Gamma \vdash \Delta, B \quad B, \Gamma \vdash \Delta, A}{\Gamma \vdash \Delta, A \leftrightarrow B} \leftrightarrow_r \quad \frac{\Gamma \vdash \Delta, A, B \quad A, B, \Gamma \vdash \Delta}{A \leftrightarrow B, \Gamma \vdash \Delta} \leftrightarrow_l$$

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$$\frac{A, \Gamma \vdash \Delta, B \quad B, \Gamma \vdash \Delta, A}{\Gamma \vdash \Delta, A \leftrightarrow B} \leftrightarrow_r \quad \frac{\Gamma \vdash \Delta, A, B \quad A, B, \Gamma \vdash \Delta}{A \leftrightarrow B, \Gamma \vdash \Delta} \leftrightarrow_l$$

The resulting calculus is:

- ▶ sound

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replace $\leftrightarrow$ everywhere it occurs with $(. \rightarrow .) \wedge (. \rightarrow .)$

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The resulting calculus is:

- ▶ sound
replace $\leftrightarrow$ everywhere it occurs with $(. \rightarrow .) \wedge (. \rightarrow .)$
- ▶ complete

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for the same reason

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The resulting calculus is:

- ▶ sound
replace $\leftrightarrow$ everywhere it occurs with $(. \rightarrow .) \wedge (. \rightarrow .)$
- ▶ complete
for the same reason
- ▶ enjoys cut-elimination

Macros for quantifiers

A macro for quantifiers M is a formula based on quantifiers $Q_i \in \{\forall, \exists\}$, $1 \leq i \leq n$ which is considered as a connective in its own right:

$$M_{x_1, \dots, x_n} A(x_1, \dots, x_n) = Q_1 x_1, \dots, Q_n x_n A(x_1, \dots, x_n).$$

Example

The quantifier macro Q :

$$Q_{x,y} A(x, y) = \forall x \exists y A(x, y).$$

The calculus **LQ**: a first approach

$$Q_{x,y}A(x,y) = \forall x \exists y A(x,y)$$

Q_r is defined via

$$\frac{\frac{\vdash A(\textcolor{red}{a}, t)}{\vdash \exists y A(\textcolor{red}{a}, y)} \exists_r}{\vdash \forall x \exists y A(\textcolor{red}{x}, y)} \forall_r$$

where a does not occur in the lower sequent.

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where a does not occur in the lower sequent. Consequently, the inference rule for Q_r is

$$\frac{\Gamma \vdash \Delta, A(\textcolor{red}{a}, t)}{\Gamma \vdash \Delta, Q_{x,y}A(x,y)} Q_r$$

where a does not occur in the lower sequent.

The calculus **LQ**: a first approach

Q_I is defined via

$$\frac{\frac{A(t, \textcolor{red}{a}) \vdash}{\textcolor{red}{\exists} y A(t, \textcolor{red}{y}) \vdash} \textcolor{red}{\exists}_I}{\forall x \textcolor{red}{\exists} y A(x, \textcolor{red}{y}) \vdash} \forall_I$$

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where a does not occur in the lower sequent. Consequently, the inference rule for Q_I is

$$\frac{A(t, \textcolor{red}{a}), \Gamma \vdash \Delta}{Q_{x,y} A(x, y), \Gamma \vdash \Delta} Q_I$$

where a does not occur in the lower sequent and in t .

The calculus **LQ**: a first approach

The calculus **LQ** is **LK** extended with

$$\frac{\Gamma \vdash \Delta, A(\textcolor{red}{a}, t)}{\Gamma \vdash \Delta, Q_{x,y}A(x, y)} Q_r \qquad \frac{A(t', \textcolor{red}{a}'), \Gamma \vdash \Delta}{Q_{x,y}A(x, y), \Gamma \vdash \Delta} Q_l$$

where a does not occur in the lower sequent and a' does not occur in the lower sequent and in t' .

The calculus **LQ**: a first approach

Example

The sequent $Q_{x,y}A(x,y) \vdash \forall x \exists y A(x,y)$ is derivable in **LQ**:

$$\frac{\frac{\frac{A(a, \textcolor{red}{b}) \vdash A(\textcolor{red}{a}, b)}{A(a, \textcolor{red}{b}) \vdash \exists y A(\textcolor{red}{a}, y)} \exists_r}{Q_{x,y}A(x,y) \vdash \exists y A(\textcolor{red}{a}, y)} Q_l}{Q_{x,y}A(x,y) \vdash \forall x \exists y A(\textcolor{red}{x}, y)} \forall_r$$

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► **LQ** is sound.

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- **LQ** is sound.

The macro Q can be replaced by $\forall \exists$ everywhere in the derivation.
The resulting derivation is an **LK**-derivation.

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- **LQ** is sound.

The macro Q can be replaced by $\forall \exists$ everywhere in the derivation.
The resulting derivation is an **LK**-derivation.

- **LQ** admits cut-elimination.

The calculus **LQ**: a first approach

- ▶ How about completeness?

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LQ is incomplete w.r.t. the sequents provable in **LK**.

The calculus **LQ**: a first approach

- How about completeness?

LQ is incomplete w.r.t. the sequents provable in **LK**.

Assume by contradiction the sequent

$$Q_{x,y}A(x,y) \vdash Q_{x,y}(A(x,y) \vee C)$$

is provable. A derivation has the initial form

$$\frac{A(a,b) \vdash A(a,b)}{A(a,b) \vdash A(a,b) \vee C}$$
$$\vdots$$

Due to the mixture of strong (eigenvariable dependent) and weak positions in Q none of the inference rules Q_r , Q_l can be applied.

The problem with **LQ**

The inherent incompleteness of **LQ** for trivial statements is a consequence of the fact that Q represents a quantifier inference macro combining a strong and a weak quantifier.

Solution: consider sequent calculi with concepts of proof which are globally but not locally sound.

This means that all derived statements are true but that not every subderivation is meaningful.

Unsound reasoning...

... but of correct results.

- (1) That Kurt Gödel is Austrian entails that Kurt Gödel is Austrian.

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- (2) Hence, that Kurt Gödel is Austrian entails that everyone is Austrian.

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- (3) That is, if Kurt Gödel is Austrian, then all people are Austrian.

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- (4) Therefore, there exists a person such that, if that person is Austrian, then all people are Austrian.

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- (4) Therefore, there exists a person such that, if that person is Austrian, then all people are Austrian.

$$\frac{\frac{\frac{A(a) \vdash A(a)}{A(a) \vdash \forall y A(y)}}{\vdash A(a) \rightarrow \forall y A(y)}}{\vdash \exists x (A(x) \rightarrow \forall y A(y))}$$

The calculus $\mathbf{LQ}^{++}$

side variable relation $<_{\varphi, \mathbf{LQ}}$

Let φ be an $\mathbf{LQ}$ -derivation. We say b is a side variable of a in φ (written $a < b$) if φ contains a strong quantifier inference of the form

$$\frac{A(t, a), \Gamma \vdash \Delta}{Q_{x,y} A(x, y), \Gamma \vdash \Delta} Q_I$$

and b occurs in t .

The calculus $\mathbf{LQ}^{++}$

$\mathbf{LQ}$ -suitable quantifier inferences

A quantifier inference is $\mathbf{LQ}$ -suitable for a proof φ if the following three conditions are satisfied:

- ▶ (substitutability) the eigenvariable does not appear in the end-sequent of φ .
- ▶ (side variable condition) the relation $<$ is acyclic.
- ▶ (weak regularity) inferences with different main formulas have different eigenvariables.

sequent calculus $\mathbf{LQ}^{++}$

The sequent calculus $\mathbf{LQ}^{++}$ is $\mathbf{LQ}$, except that we replace quantifier inferences with $\mathbf{LQ}$ -suitable quantifier inferences.

The calculus $\mathbf{LQ}^{++}$

Example

The sequent $Q_{x,y}A(x,y) \vdash Q_{x,y}(A(x,y) \vee C)$ is $\mathbf{LQ}^{++}$ -derivable.
Consider the derivation $\varphi =$

$$\frac{\frac{\frac{A(a,b) \vdash A(a,b)}{A(a,b) \vdash A(a,b) \vee C} w_r + \vee_r}{A(a,b) \vdash Q_{x,y}(A(x,y) \vee C)} Q_r}{Q_{x,y}A(x,y) \vdash Q_{x,y}(A(x,y) \vee C)} Q_I$$

with $b < a$.

Just for fun?

Yes, but...

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“Some relative of each villager and some relative of each townsman hate each other.”

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Yes, but...

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- ▶ branched quantifiers, finite partially ordered quantifiers or nonlinear quantifiers.

The simplest Henkin quantifier (not definable in ordinary first-order logic) is Q_H :

$$A_H = \left(\begin{array}{cc} \forall x & \exists u \\ \forall y & \exists v \end{array} \right) A(x, y, u, v).$$

For every x there is a u and for every y there is a v (depending only on y) such that $A(x, y, u, v)$.

Why this is not cheating

Two forms of unsoundness:

- ▶ arbitrary shortcuts $\rightarrow$ no recoverability.
- ▶ controlled shortcuts $\rightarrow$ externally checkable.

Known paradigms:

- ▶ abstraction in model checking
- ▶ heuristic SAT solving
- ▶ machine-assisted theorem proving (where candidates are wrong until checked)

Unsound reasoning is acceptable only when it is embedded in a system that can detect or correct its failures.

Final takeaway

Unsound ideas are not the problem, unmonitored unsoundness is.

In research practice, we do something similar: early ideas are often locally incorrect or incomplete, but become meaningful once embedded in the right formal or social structure – peer review, refinement, and proof reconstruction.